

Order Flow Imbalance (OFI) - Conceptual Questions

Shreya Arya

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1 Question 1

What's the motivation behind measuring OFI at multiple depth levels of the order book?

Measuring OFI at multiple depth levels gives a more insight into the complete picture of the market's buying and selling pressure. While the best bid and ask prices (Level 0) reflect immediate trading interest, deeper levels can reveal hidden liquidity and early signs of pressure that haven't yet surfaced at the top of the book. For instance, large institutional traders often split their orders and place them across several price levels to avoid moving the market. By looking solely at the top level, we may miss this deeper activity. Measuring OFI across multiple levels helps capture these larger strategic flows, making it a more robust signal for predicting short-term price movements.

2 Question 2

Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors use LASSO regression instead of OLS to deal with the high-dimensional and sparse nature of cross-asset order flow data. When modeling how OFI in one asset affects the price or flow in many other assets, there may be dozens or hundreds of potential predictors, but only a few are truly relevant. LASSO adds a regularization penalty that shrinks unimportant coefficients to zero, effectively selecting only the most impactful variables. This reduces overfitting, improves model interpretability, and allows for automatic feature selection — which is especially useful in financial markets where relationships are often noisy and unstable.

3 Question 3

Why is OFI considered a better predictor of short-term returns than trade volume?

OFI is more informative than trade volume because it captures the directional pressure in the market, not just activity. Trade volume only tells you how many shares changed hands — it doesn't tell you whether buyers or sellers were in control. On the other hand, OFI looks at how limit orders are added, canceled, or executed at each level of the order book. It reflects whether buyers are becoming more aggressive (adding bids, removing asks) or if sellers are (adding asks, removing bids). Since prices move based on supply and demand imbalance, OFI provides a direct signal of market pressure, making it a stronger predictor of near-term price changes than raw volume alone.